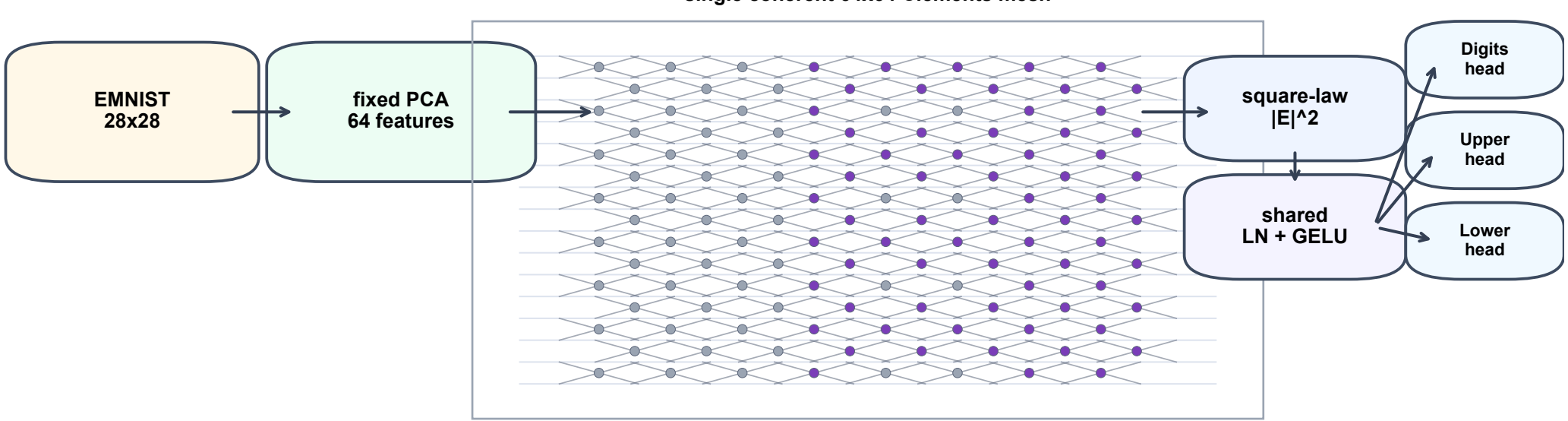
Single-Mesh Multi-Task Photonic Network

Shared MPU pattern is learned during design: gray shared neurons use common theta/phi across tasks; purple private neurons keep task-specific phase deltas.



64 optical modes, alternating even/odd 2x2 stages; one final detection

Implementation note: code uses one SharedClementsMZILayer with theta_even/phi_even in R^(32x32) and theta_odd/phi_odd in R^(32x31), total 2016 2x2 units.

shared MPU task-private MZI